

NumericalMethods

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1 Exact solution

We investigate the following equation:

$$u(t, x, y) = e^{i(k_x x + k_y y - \omega t)} \quad (1)$$

as a solution to the wave equation.

$$\frac{\partial^2 u(t, x, y)}{\partial t^2} = c^2 \nabla^2 u(t, x, y) \quad (2)$$

We investigate first the first time derivative of $u(t, x, y)$.

$$\frac{\partial u}{\partial t} = -i\omega e^{i(k_x x + k_y y - \omega t)} \quad (3)$$

Finding the second time derivative yields:

$$\frac{\partial^2 u}{\partial t^2} = -\omega^2 e^{i(k_x x + k_y y - \omega t)} = -\omega^2 u(t, x, y) \quad (4)$$

We acknowledge the similar form to $\cos(\omega t)$ and $\sin(\omega t)$.

We repeat the same steps with space derivatives. We do for x but the steps for y are identical.

$$\frac{\partial u}{\partial x} = -ik_x e^{i(k_x x + k_y y - \omega t)} \quad (5)$$

repeating again.

$$\frac{\partial^2 u}{\partial x^2} = -k_x^2 e^{i(k_x x + k_y y - \omega t)} = -k_x^2 u(t, x, y) \quad (6)$$

Reminding ourselves that in 2D $\nabla^2 u$ is defined in the following way.

$$\nabla^2 u(t, x, y) = \left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} \right) u \quad (7)$$

Inputting the derivatives as found earlier into the wave equation we are able to show that the equation in fact does satisfy the wave equation. We also remind

ourselves that $c = \frac{w}{k}$ and that $k = (k_x^2 + k_y^2)^{\frac{1}{2}}$. We start with left-hand side of the wave equation.

$$\frac{\partial^2 u(t, x, y)}{\partial t^2} = -w^2 u(t, x, y) \quad (8)$$

Moving onto the Right-Hand side of the equation.

$$c^2 \nabla^2 u(t, x, y) = \frac{w^2}{k^2} \left(\frac{\partial^2 u(t, x, y)}{\partial x^2} + \frac{\partial^2 u(t, x, y)}{\partial y^2} \right) \quad (9)$$

$$\frac{w^2}{k^2} \left(\frac{\partial^2 u(t, x, y)}{\partial x^2} + \frac{\partial^2 u(t, x, y)}{\partial y^2} \right) = \frac{w^2}{k^2} \cdot (-k_x^2 - k_y^2) u(t, x, y) \quad (10)$$

simplifying the equation yields us the Left-Hand side.

$$\frac{w^2}{k^2} \cdot (-k_x^2 - k_y^2) u(t, x, y) = -w^2 u(t, x, y) \quad (11)$$

We have thus shown that the exact solution proposed satisfies the wave equation.

2 Dispersion Coefficient

We aim to show that the discretized exact solution is exact when the courant-friedrich-lewy number is $C = \frac{1}{\sqrt{2}}$ and the following conditions $mx = my$ and $k_x = k_y = k$ are met. We first give the discretized exact solution.

$$u_{i,j}^n = e^{i(kh(i+j) - \bar{w}n\Delta t)} \quad (12)$$

We add our equation into the discretized central difference equation as given below.

$$\frac{u_{i,j}^{n+1} - 2u_{i,j}^n + u_{i,j}^{n-1}}{(\Delta t)^2} = c^2 \left(\frac{u_{i+1,j}^n - 2u_{i,j}^n + u_{i,j-1}^n}{h^2} + \frac{u_{i,j+1}^n - 2u_{i,j}^n + u_{i,j-1}^n}{h^2} \right) \quad (13)$$

We focus on the left hand side. We notice that in all cases we might simplify in a very similar manner. Below is an example of simplification for the case of $n+1$.

$$u_{i,j}^{n+1} = e^{-i\bar{w}\Delta t} u_{i,j}^n \quad (14)$$

Identically we might write for case of $i+1$ or $j+1$.

$$u_{i+1,j}^n = u_{i,j+1}^n = e^{ikh} u_{i,j}^n \quad (15)$$

Awkwardly we must note that index i is not same as i in the exponent. The i in exponents refer to the complex number i defined as $i = \sqrt{-1}$, contrarily the i in u is a spatial index of the mesh function u .

Returning back to the left hand side we simplify the expression.

$$\frac{u_{i,j}^{n+1} - 2u_{i,j}^n + u_{i,j}^{n-1}}{(\Delta t)^2} = \frac{u_{i,j}^n}{(\Delta t)^2} (e^{-i\bar{w}\Delta t} - 2 + e^{i\bar{w}\Delta t}) \quad (16)$$

Now that we have a simplified expression for left-hand side based on the discretized exact solution we move onto the right hand side. We choose to focus on the individual fractions first as to not overwhelm with algebra.

$$\frac{u_{i+1,j}^n - 2u_{i,j}^n + u_{i,j-1}^n}{h^2} = \frac{u_{i,j}^n}{h^2} (e^{ikh} - 2 + e^{-ikh}) \quad (17)$$

We additionally note that this is the same form for the variation in the j index and further highlight that the i in the exponents is not an index.

We look to combine both the right-hand and left-hand side expressions. We do so below.

$$\frac{u_{i,j}^n}{(\Delta t)^2} (e^{-i\tilde{w}\Delta t} - 2 + e^{+i\tilde{w}\Delta t}) = c^2 \frac{2u_{i,j}^n}{h^2} ((e^{ikh} - 2 + e^{-ikh})) \quad (18)$$

Recognise that c is not the courant number and is instead the wave speed. The Courant number is defined as following, $C = \frac{c\Delta t}{h}$. We combine the constants so as to get the courant number into our equation and cancel out the common factors.

$$(e^{-i\tilde{w}\Delta t} - 2 + e^{+i\tilde{w}\Delta t}) = 2C^2 (e^{ikh} - 2 + e^{-ikh}) \quad (19)$$

To make our job simpler we choose to return to the conditions. We define then $C = \frac{1}{\sqrt{2}}$. This simplifies our equation as shown below.

$$e^{-i\tilde{w}\Delta t} + e^{+i\tilde{w}\Delta t} = e^{ikh} + e^{-ikh} \quad (20)$$

To advance further into getting a relation between $\tilde{w}$ and k we recall the connection between imaginary exponents and cos. This is recalled below.

$$e^x + e^{-x} = 2\cos(x) \quad (21)$$

We apply the above relation.

$$\cos(\tilde{w}\Delta t) = \cos(kh) \quad (22)$$

We can then find a relation for $\tilde{w}$ based on the above description. We note that $\frac{h}{\Delta t} = \sqrt{2}c$.

$$\tilde{w} = \sqrt{2}kc \quad (23)$$

This condition is nice but we have no w to compare to. We understand that in general for a 2-dimensional wave w is given in the following manner.

$$w^2 = c^2(k_x^2 + k_y^2) \quad (24)$$

We remember that one of the conditions we set was $k_x = k_y = k$. Using this condition we see.

$$w^2 = 2c^2k^2 \quad (25)$$

it follows immediately that $w = \sqrt{2}ck$ which is exactly equivalent to our $\tilde{w}$. We have then shown that our discretized exact equation is equal to the exact equation when $C = \frac{1}{\sqrt{2}}$ with the additional conditions $k_x = k_y = k$ and $m_x = m_y = m$.